

## **ANATOMY - MOCK MIDTERM 3**

Head and neck: the face, the mouth and nose, and the eye

**Instructions.** 24 questions. Only **one answer** is correct for each question. Suggested time: 40 minutes. Answer on your own first, then check the answer key at the end, which explains each answer and names the trap the question uses. This paper was written for practice by Sina and is not an official exam.

**Question 1/24 Which anatomical parts form the floor of the orbit?**

- A. The orbital plate of the frontal bone
- B. The lacrimal bone and the ethmoid bone
- C. The orbital surface of the maxillary bone, the zygomatic bone and the orbital process of the palatine bone
- D. The greater wing of the sphenoid bone

**Question 2/24 The medial wall of the orbit:**

- A. Is formed by the greater wing of the sphenoid bone
- B. Is the strongest wall of the orbit
- C. Is formed mainly by the lamina papyracea of the ethmoid bone
- D. Contains the fossa for the lacrimal gland

**Question 3/24 The optic canal:**

- A. Joins the orbit to the pterygopalatine fossa
- B. Transmits the ophthalmic veins
- C. Lies between the greater and the lesser wings of the sphenoid bone
- D. Transmits the optic nerve and the ophthalmic artery

**Question 4/24 The inferior orbital fissure:**

- A. Joins the orbit to the middle cranial fossa
- B. Transmits the infraorbital nerve
- C. Lies between the lesser and the greater wings of the sphenoid bone
- D. Transmits the oculomotor nerve

**Question 5/24 The palatine bone:**

- A. Is L-shaped, with a horizontal and a perpendicular plate
- B. Forms the anterior part of the hard palate
- C. Forms the bridge of the nose
- D. Is an unpaired bone

**Question 6/24 Regarding the skull:**

- A. It is formed of 22 bones: 8 cranial and 14 facial
- B. The vomer is a paired bone
- C. The frontal bone is a facial bone
- D. The mandible articulates with the zygomatic bone

**Question 7/24 The frontal sinus:**

- A. Is supplied by the maxillary nerve
- B. Is well developed at birth
- C. Drains into the middle meatus through the frontonasal duct
- D. Lies on the sides of the nasal cavity

**Question 8/24 The choanae:**

- A. Open into the oral cavity
- B. Are separated by the perpendicular plate of the ethmoid bone
- C. Open the nasal cavities into the nasopharynx
- D. Are the nostrils

**Question 9/24 Regarding the tongue:**

- A. Its intrinsic muscles are attached to the hyoid bone
- B. General sensation of its posterior 1/3 is carried by the lingual nerve
- C. Retraction is produced by the genioglossus muscle
- D. Lymph from its tip drains to the submental nodes on both sides

**Question 10/24 Secretomotor fibers to the parotid gland synapse in:**

- A. The submandibular ganglion
- B. The otic ganglion
- C. The pterygopalatine ganglion
- D. The ciliary ganglion

**Question 11/24 The facial nerve:**

- A. Supplies the muscles of mastication
- B. Carries the general sensation of the face
- C. Divides into its terminal branches within the parotid gland
- D. Leaves the cranial cavity through the jugular foramen

**Question 12/24 A patient cannot close the right eye nor raise the right eyebrow, and the right corner of the mouth droops. The lesion most likely involves:**

- A. The right oculomotor nerve
- B. The left facial nerve
- C. The right trigeminal nerve
- D. The right facial nerve

**Question 13/24 Regarding the temporalis muscle:**

- A. It is supplied by the facial nerve
- B. It inserts on the angle of the mandible
- C. It protrudes the mandible
- D. Its posterior fibers retract the mandible

**Question 14/24 Temporo-mandibular joint:**

- A. The parotid gland lies medial to the joint
- B. Its lateral ligament prevents anterior dislocation
- C. Elevation and depression occur in its upper cavity
- D. Protrusion and retraction occur in its upper cavity

**Question 15/24 The lingual artery:**

- A. Arises from the internal carotid artery
- B. Is a branch of the facial artery
- C. Ends as the angular artery
- D. Is divided into three parts by the hyoglossus muscle

**Question 16/24 The retromandibular vein:**

- A. Drains directly into the subclavian vein
- B. Is formed by the union of the superficial temporal and maxillary veins
- C. Is formed within the submandibular gland
- D. Is formed by the angular and facial veins

**Question 17/24 Within the parotid gland, from superficial to deep, we find:**

- A. The facial nerve, the retromandibular vein, the external carotid artery
- B. The external carotid artery, the retromandibular vein, the facial nerve
- C. The facial nerve, the external carotid artery, the retromandibular vein
- D. The retromandibular vein, the facial nerve, the external carotid artery

**Question 18/24 The levator palpebrae superioris:**

- A. Arises from the floor of the orbit
- B. Arises from the lesser wing of the sphenoid bone
- C. Is supplied by the facial nerve
- D. Closes the eyelid

**Question 19/24 Tears drain from the lacrimal sac through the nasolacrimal duct into:**

- A. The nasopharynx
- B. The maxillary sinus
- C. The middle meatus
- D. The inferior meatus

**Question 20/24 The uvea:**

- A. Is formed by the iris, the ciliary body and the choroid
- B. Is the outer coat of the eyeball
- C. Includes the retina
- D. Includes the sclera

**Question 21/24 Which of the following are responsible for generating action potentials in the retina?**

- A. Ganglion cells
- B. Photoreceptors
- C. Bipolar cells
- D. Horizontal cells

**Question 22/24 The inferior oblique muscle:**

- A. Elevates and abducts the eye
- B. Arises from the common tendinous ring
- C. Is supplied by the trochlear nerve
- D. Inserts deep to the superior rectus muscle

**Question 23/24 Light shone into the right eye produces no constriction of either pupil, while light shone into the left eye constricts both pupils. The lesion is in the:**

- A. Left oculomotor nerve
- B. Left optic nerve
- C. Right optic nerve
- D. Right oculomotor nerve

**Question 24/24** When the cornea of one eye is touched, both eyes blink. The afferent limb of this reflex is:

- A.** The facial nerve
- B.** The ophthalmic nerve (V1)
- C.** The oculomotor nerve
- D.** The optic nerve

# Answer key, mock midterm 3

| Q  | Ans      | Why                                                                                                                                      | Trap used                                                           |
|----|----------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1  | <b>C</b> | The floor is also the roof of the maxillary sinus.                                                                                       | -                                                                   |
| 2  | <b>C</b> | Medial wall = frontal process of maxilla, lacrimal, ethmoid, body of sphenoid; thinnest wall. The lacrimal gland fossa is superolateral. | Strongest/thinnest; gland fossa vs sac fossa.                       |
| 3  | <b>D</b> | It is in the lesser wing; the veins use the superior orbital fissure.                                                                    | Canal vs fissure.                                                   |
| 4  | <b>B</b> | Between the greater wing and the maxilla; joins the orbit to the pterygopalatine fossa; V2 branches.                                     | Superior/inferior fissure swap.                                     |
| 5  | <b>A</b> | Paired; its horizontal plate forms the posterior hard palate.                                                                            | Anterior/posterior swap.                                            |
| 6  | <b>A</b> | Unpaired facial bones: mandible and vomer; the mandible articulates only with the temporal bone.                                         | Numbers swapped in other papers (14 cranial, 8 facial).             |
| 7  | <b>C</b> | Rudimentary at birth; V1; above the orbits behind the brows.                                                                             | 'Sides of the nasal cavity' is a real-paper distractor.             |
| 8  | <b>C</b> | They are separated by the vomer.                                                                                                         | -                                                                   |
| 9  | <b>D</b> | Genioglossus protrudes; posterior 1/3 = IX; intrinsic muscles have no bony attachment.                                                   | Real-paper idea ('lymph can reach bilateral nodes').                |
| 10 | <b>B</b> | IX → lesser petrosal → otic ganglion → auriculotemporal (V3).                                                                            | Four parasympathetic ganglia.                                       |
| 11 | <b>C</b> | Stylomastoid foramen; muscles of facial expression; sensation of the face is V.                                                          | Real-paper stem, distractors from the 2023-24 midterm.              |
| 12 | <b>D</b> | A lower motor neuron lesion of VII affects the whole half of the face on the same side.                                                  | Side and nerve swaps.                                               |
| 13 | <b>D</b> | Inserts on the coronoid process; elevates; V3.                                                                                           | 'The temporalis protrudes the mandible' is a real-paper distractor. |
| 14 | <b>D</b> | Lower cavity = hinge (elevation/depression); the lateral ligament prevents posterior dislocation; parotid is lateral.                    | Upper/lower and anterior/posterior swaps.                           |
| 15 | <b>D</b> | Branch of the ECA opposite the tip of the greater horn of the hyoid.                                                                     | Lingual vs facial.                                                  |
| 16 | <b>B</b> | Formed inside the parotid; anterior division → facial vein, posterior division → external jugular.                                       | Gland swap.                                                         |
| 17 | <b>A</b> | Nerve, vein, artery from outside in.                                                                                                     | Order question.                                                     |
| 18 | <b>B</b> | CN III (its smooth part is sympathetic); raises the lid; the inferior oblique arises from the floor.                                     | III/VII swap.                                                       |
| 19 | <b>D</b> | The duct opens under the inferior concha.                                                                                                | Meatus swap (the real midterm used 'superior').                     |
| 20 | <b>A</b> | Middle (vascular) coat.                                                                                                                  | -                                                                   |
| 21 | <b>A</b> | Photoreceptors convert light; photoreceptors and bipolar cells make graded potentials; ganglion axons form the optic nerve.              | Mirror of the real mock question ('converting light').              |
| 22 | <b>A</b> | Inferior branch of III; from the anterior floor; inserts deep to the lateral rectus; elevation, abduction, extorsion.                    | Superior/inferior oblique swaps.                                    |

| <b>Q</b> | <b>Ans</b> | <b>Why</b>                                                                | <b>Trap used</b>                                 |
|----------|------------|---------------------------------------------------------------------------|--------------------------------------------------|
| 23       | <b>C</b>   | The right eye cannot send the signal in (afferent), but both CN III work. | Afferent vs efferent reasoning.                  |
| 24       | <b>B</b>   | V1 → spinal nucleus of V → both facial nuclei → VII to orbicularis oculi. | Stem explains the reflex, as the professor does. |